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AI-generated tactile graphics for visually impaired children: A usability study of a multimodal educational product

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ABSTRACT

Approximately 70 million children aged 0–14 worldwide have visual impairments, limiting their language acquisition and image recognition development due to a lack of visual input. Human-computer interaction technologies provide an opportunity to learn Braille and images through touch and auditory stimuli, replacing traditional visual input elements. However, effectively integrating these sensory inputs remains a challenge. To address this, we developed "TaleVision," a product that uses large language models to instantly generate touchable images and sound effects, allowing visually impaired children to better understand the connection between Braille and images. We recruited five sighted children and 14 children with congenital visual impairments to test our product. Almost all participants reported high satisfaction and rated the user experience highly. Additionally, we also observed how visually impaired children and teachers interacted with our products to improve teaching and learning. Based on the results, we discussed best design practices, process recommendations, and potential future solutions.

Additional Keywords and Phrases

Visual Impairments Children, Large Language Model, DALLE-3, Cognitive Education, Multimodal Perception

1. Introduction

Globally, approximately 70 million children aged 0–14 experience visual impairments (Yekta et al., 2022; Bourne et al., 2021). While some retain light perception and can receive limited visual information, the lack of visual input poses significant challenges to their language learning and cognitive development. These children often struggle to associate the shape, outline, and details of objects with the vocabulary and sentences they are taught. Traditional teaching aids for visually impaired children exhibit notable limitations (Chokron et al., 2020; Rogge et al., 2021). Firstly, the interactive methods required by current teaching tools for visually impaired children are labor-intensive, with teachers manually pressing Braille, tactile writing system of raised dots that represent literacy and information, and presenting it to students, and presenting it to students (Niwagaba, 2014). Secondly, some tools merely pair Braille with everyday objects for touch-based learning, limiting the ability to convey larger and more complex visual concepts (Permatasari, 2022). As a result, specialized educational methods and assistive tools are necessary to help visually impaired children overcome cognitive developmental barriers.

Technological innovations driven by human-computer interaction have the potential to revolutionize education for visually impaired children by offering tactile-based learning opportunities (Mascle et al., 2022a; Cuturi et al., 2016). For instance, tactile tablet devices with pin arrays allow users to create multiple illustrations, or even entire books, on a single platform (Mascle et al., 2022b). Early and frequent exposure to tactile materials has been shown to enhance tactile abilities, making these devices particularly effective for educational purposes (Theurel et al., 2013). Although tactile tablets are limited in their capacity to display complex images, their ability to represent simplified figures makes them well-suited for visually impaired learners. Furthermore, 80 % of the information humans acquire from their surroundings is perceived through vision (Zhou et al., 2023; Deng et al., 2020; Duchowski, 2017). Therefore, for children who have lost their sight, relying on multisensory input can be more effective in helping them understand the characteristics of objects (Tanhua-Piiroinen et al., 2008; Sánchez et al., 2010).

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However, existing tactile devices present several limitations that restrict their educational potential. Most tactile tablets are expensive and difficult to scale for widespread use. Additionally, these devices often rely on pre-designed tactile graphics, lacking the ability to generate real-time, customized images. This absence of real-time interactivity prevents children from exploring new concepts spontaneously, limiting their learning autonomy. Moreover, the current solutions are not portable, making it difficult for visually impaired children to engage in learning outside the classroom. To address these issues, an increasing number of interactive products integrate auditory modes, allowing children to comprehend objects more holistically by receiving information through multiple senses (Liu et al., 2022). A notable example is a multi-sensory interactive book designed for visually impaired children, which produces corresponding sounds when objects on the interactive device are touched (Edirisinghe et al., 2022). By incorporating auditory and tactile inputs, this approach enriches the learning experience and deepens object comprehension (Plimmer et al., 2011).

Despite these advances, there is still a significant gap in developing low-cost, portable, and real-time interactive tools that seamlessly integrate speech, tactile feedback, and auditory cues. Such systems would not only reduce the burden on educators but also empower visually impaired children to explore independently without constant supervision. Producing tactile images remains time-consuming, requires specialized skills, and tactile books tend to be expensive. Current interactive products are limited in the quantity of images they can display, often requiring manual design and editing, which reduces their reusability. Moreover, while the combination of auditory and tactile inputs is advantageous for cognitive development, integrating these sensory modalities in a seamless and effective manner still demands further technological innovation.

To address these issues, we developed TaleVision, an educational tool that leverages multimodal sensory integration and large language models to support Braille learning and language cognition for visually impaired children. TaleVision instantly generates touchable images and sound effects based on voice input, enabling children to better understand the relationship between Braille and images. This real-time generation capability distinguishes TaleVision from conventional tactile devices, which rely on pre-made graphics and cannot adapt to spontaneous learning needs. By combining tactile and auditory inputs, TaleVision offers an engaging and effective learning experience. Sensors generate raised image contours and textures based on voice-input descriptions, presenting visually impaired children with an immersive and interactive learning tool. This low-cost and portable system empowers both independent learning and teacher-assisted education, addressing the limitations of traditional tactile devices.

We assessed the accuracy of the generated images with a group of 5 sighted children, followed by a usability study with 14 children with congenital visual impairments. We recorded the accuracy rates of sighted participants, interactive behavior and failure causes during child-teacher sessions, children's satisfaction, and their user experience data. This study examines whether visually impaired children accept and how they interact with the product, addressing the following research questions:

RQ1: Can large language models serve as a technological foundation for educational tools aimed at enhancing graphical cognition in visually impaired children?
RQ2: How do visually impaired children collaborate with teachers while using the product, and what is the manifestations of their interaction?
RQ3: What are the experiences, perceptions, and preferences of visually impaired children and their teachers regarding the use of multimodal image cognition products, particularly with respect to this form of interaction?

2. Related work

2.1. Educational products for visually impaired children

In the context of blind education, teaching aids primarily serve to concretize abstract instructional content, enabling visually impaired children to comprehend complex theoretical concepts through tactile interaction. Scholars have explored the application and impact of sensory experiences in toy design, particularly in cultivating the perceptual abilities of visually impaired children. Despite some innovations in toy design, strategies for integrating tactile and auditory sensory elements remain largely theoretical, with practical implementations often focusing on enhancing texture to represent object characteristics. Early education for visually impaired children typically encompasses reading and writing, daily skill training, and orientation and mobility. Through research, we identified three main categories of existing educational products. The first consists of materials that directly assist visually impaired children in learning, such as Braille books and illustrated texts. The second includes physical educational toys with Braille markers, such as shape cards. The third category integrates smart technology into early educational tools, such as audio books and interactive toys (Nazari et al., 2023).

Smart technology has seen preliminary advances in the education of visually impaired children (Maćkowski et al., 2022). Through speech synthesis and recognition, children can access auditory information, which has been widely applied in reading, writing, and dictation, providing a more convenient learning experience (Radecki et al., 2020; Shoaib et al., 2023). Screen reading software converts text into speech or Braille, enabling children to access written content through auditory or tactile means (Maćkowski et al., 2023). These programs typically allow customization, such as adjusting speech rate and text size, to meet diverse learning needs. Computer-assisted teaching tools also offer interactive opportunities, including Braille typing software, mathematical formula editors, and educational games, which facilitate a deeper understanding of academic content (Morrison et al., 2021; Fernando and Ohene-Djan 2022). Additionally, smart devices like wristbands and glasses offer visually impaired children enhanced daily convenience and safety (Hu et al., 2022; Rao et al., 2021). Equipped with features such as navigation, speech recognition, and real-time positioning, these tools enable better social integration (Hersh, 2020; Façanha et al., 2020).

Among the advancements in tactile display technology, the APH Monarch and Dot Pad stand out as significant innovations, though they each present limitations that impact their usability for younger children. The APH Monarch, developed by the American Printing House, is a tactile device designed to display Braille text and graphics (Zakrajsek et al., 2024). While it provides a high level of functionality, its price, which reaches several thousand dollars, makes it inaccessible to many educational institutions. Additionally, the Monarch relies on pre-existing Braille books, limiting its adaptability to new content. The Dot Pad, developed by Dot Inc., uses a grid of 2400 micro pins to display tactile images and Braille, allowing users to interact with both visual and textual content. However, it requires pre-existing visual input to generate images, which is not ideal for younger children who rely on voice input to interact with educational content. These limitations highlight the need for more inclusive, flexible tactile devices that can accommodate the unique needs of younger, voice-driven learners.

Despite these advancements, significant challenges remain, such as high costs, limited user experience, and low technology adoption rates. Future research should focus on improving accessibility, reducing costs, and optimizing user experience to enhance the quality and effectiveness of education for visually impaired children (Mascle et al., 2022a; Slade et al., 2021). Moreover, the development of new technologies should prioritize sensory integration to support more comprehensive cognitive development for visually impaired students (Fig. 1).



Fig. 1. Tools of cognition for traditionally visually impaired children.

2.2. Haptic display technology

Tactile images play a crucial role in the education and cognitive development of visually impaired children, particularly in the absence of visual input. Effectively conveying the shape and properties of objects is essential in this context. Studies have shown that blindfolded children can only recognize common objects (such as bottles, umbrellas, and keys) with an accuracy of 9 %, whereas visually impaired children aged 8 to 14 exhibit a significantly higher recognition rate of 45 % (D'Angiulli et al., 1998). This highlights the distinct advantage of tactile image design. However, recognizing objects from tactile images, especially two-dimensional representations, remains challenging (Theurel et al., 2013). Given that this is one of the primary methods for describing the world to visually impaired individuals, optimizing tactile image design to improve object recognition is a key research focus.

Currently, the most commonly used devices among visually impaired children are tactile displays, which allow users to read various content through touch by interacting with dynamic dot arrays (Kintonova et al., 2024). Tactile display stimuli are primarily categorized into thermal, electrical, and mechanical. Due to the poor perceptual accuracy and discomfort associated with thermal and electrical stimuli, mechanical stimuli have become the dominant choice for long-term use. Mechanical stimuli simulate tactile experiences through different driving technologies, including piezoelectric ceramics, electromagnetic actuators, shape-memory alloys, and pneumatic drives (Germani et al., 2013; Shi et al., 2021). Piezoelectric ceramics generate small electrical pulses through voltage changes, shape-memory alloys provide tactile feedback by deforming materials, and pneumatic drives regulate the tension of the touch layer through airflow control (Liu and Xu, 2014). The introduction of new composite materials, such as nanoparticle-enhanced materials, has further improved the precision and comfort of tactile experiences.

Piezoelectric ceramic actuators are widely employed in tactile displays, where each tactile point corresponds to a driving unit. Although this technology offers fast response times, the limited stretchability of piezoelectric ceramics makes it difficult to meet the portability requirements of devices. Other driving technologies also present challenges: shape-memory alloys, while capable of high contact point density, suffer from slow response times and fatigue; electromagnetic actuators, although responsive and cost-effective, face issues with power consumption and heat generation; and pneumatic drives are hindered by large device size and complex structures (Pai and Hu, 2023; Wai and Lee, 2004).

In summary, understanding whether appropriate representation technologies can enhance object recognition and evaluating the extent to which these technologies are applicable to all visually impaired individuals are of significant research interest (Theurel et al., 2013).

2.3. Large language model

Artificial intelligence (AI) has fundamentally transformed the education system. One notable application is AI's ability to convert text into images, which has diverse implications across various fields. For instance, AI can automatically generate film storyboards by converting scripts into visual sequences or transform textual narratives into webcomic formats. It can also turn product descriptions from plain text into visually engaging, easy-to-understand images (Lee et al., 2024). Due to the wide range of applications, numerous studies have been conducted on text-to-image generation. Prominent approaches include methods based on Generative Adversarial Networks (GANs), autoregressive models, and diffusion models. GANs use a dual-network structure where a generator produces images and a discriminator evaluates their authenticity, enabling the system to improve through continuous competition. Autoregressive models generate data sequentially by predicting each component based on prior outputs, ensuring coherent and context-aware generation. Diffusion models progressively refine random noise into clear images through a reverse diffusion process, allowing for the creation of high-quality, detailed visual representations. Each of these approaches contributes to enhancing the accuracy and diversity of generated outputs, offering distinct advantages for various multimodal applications. Although the outcomes of these methods are not yet flawless, many users have expressed satisfaction with the generated images (Kim et al., 2024). Among these systems, OpenAI's DALL-E is particularly noteworthy (Gozalo-Brizuela and Garrido-Merchan, 2023).

DALL-E utilizes the Transformer architecture, which has demonstrated exceptional performance in natural language processing tasks (Ding et al., 2021). The model is trained on vast amounts of text-image pairings, learning to associate textual descriptions with corresponding visual elements. During training, the model acquires the ability to match vocabulary in text with visual features. When a user inputs text, DALL-E encodes it into semantic vectors and gradually constructs an image, starting with rough outlines and progressing to detailed elements. It then uses the VQ-VAE encoder, which is a generative model that compresses input data into a discrete space using vector quantization, to decode and ultimately generate high-resolution images (Maerten and Soydaner, 2023). The generation process incorporates a degree of randomness, allowing DALL-E to produce images that are both diverse and creative, even generating unprecedented, imaginative combinations (Vayadande et al., 2023). Its self-supervised learning framework enables the model to enhance its understanding of the relationship between image details and semantic associations by analyzing large-scale text-image pairs, significantly improving its image generation capabilities (Fig. 2).

Given these attributes, DALL-E holds substantial potential in educational settings, particularly in creating tactile images for visually impaired children (Maharana et al., 2022). By transforming complex visual information into Braille and tactile images, DALL-E can facilitate



Fig. 2. Pixelated image generated by DALL-E (not fine-tuned).

more intuitive comprehension of abstract concepts, addressing the limitations of current tactile teaching aids and substantially enriching the learning experience for visually impaired students.

3. Design

This study aims to assess the usability of a multimodal perception integration product, based on large language models, for visually impaired children's image cognition. This section details the development of the TaleVision product designed for this purpose. We first provide an overview of stakeholders' needs, followed by a description of the interaction framework, hardware and software development.

3.1. Stakeholder research

To understand the interactive needs of visually impaired children in image-based education, the product adheres to a user-centered design methodology (Abras et al., 2004; Sanders, 2002), incorporating extensive input from blind adults, teachers at schools for the visually impaired, and visually impaired children themselves. Over four months, three researchers collected design requirements (Table 1).

Table 1
Pain points and demands of different stakeholders.

Stakeholder	Pain Points	Demands
Visually Impaired Children	– Limited ability to associate Braille with real-world objects and concepts. – Braille learning is time-consuming and sometimes tedious, leading to low motivation.	– Multisensory learning tools that integrate tactile, auditory, and real-time feedback. – A more engaging and interactive learning process.
Blind Adults (Reflecting on Childhood Learning Experiences)	– Difficulty understanding large-scale, untouchable objects (e.g., "mountains," "cloud"). – Current tactile perception devices are not portable, making it difficult for users to engage in learning outside the home or classroom.	– Enhanced accessibility to a broader range of visual concepts. – Intuitive interfaces specifically designed for blind users to facilitate independent learning.
Teachers at Schools for the Visually Impaired	– Existing teaching aids are labor-intensive to create and maintain. – Manual production of Braille materials is time-consuming and inefficient.	– Intelligent, scalable teaching tools to support classroom learning. – Solutions that reduce manual workload while improving teaching efficiency.
Parents of Visually Impaired Children	– Limited access to engaging at-home learning materials. – Struggles with guiding Braille learning due to a lack of specialized training. – Concerns about their child's independence and long-term learning motivation.	– A real-time interactive learning system that allows children to explore independently without constant supervision. – An intuitive, easy-to-use interactive system that enables parents to assist their children's learning more easily.

To address the potential difficulties visually impaired children face in articulating their learning challenges, both blind adults and visually impaired children were involved in the study, with the adults reflecting on the difficulties they encountered in their childhood learning. Due to their lack of visual input, visually impaired children rely solely on tactile perception to recognize the shape of Braille dots, limiting their comprehensive understanding of text and its meaning. While most visually impaired children retain their auditory and cognitive abilities, the absence of visual cues for large, untouchable objects, such as "mountains" and "trees," complicates their image cognition. Braille learning typically requires significant time and effort, and the inability to effectively associate text with real-world objects and concepts hampers learning efficiency and comprehension. This prolonged learning process often leads to boredom and a loss of interest, further exacerbating low learning efficiency. Improving the Braille learning experience and enhancing the learning process for visually impaired children thus became a key motivation for designing innovative learning tools.

Feedback from teachers also revealed numerous challenges in Braille instruction. Current teaching aids, often manually crafted by teachers, help visually impaired children to some extent, but they are time-consuming to produce, difficult to scale, and insufficient to meet classroom demands. Furthermore, the tactile learning materials available in schools fail to fully represent the appearance of objects, leading students to misunderstand the content. This not only affects teaching efficiency but also increases teachers' workload. To support better classroom instruction and Braille lesson preparation, there is an urgent need to develop intelligent, user-friendly teaching aids that simplify the Braille teaching process and improve educational outcomes.

Based on the needs of the visually impaired and their teachers, this study seeks to design a multimodal learning tool that integrates tactile, auditory, and visual perception. By combining these sensory modalities, the tool aims to help visually impaired children learn Braille more efficiently. The integration of different sensory inputs provides them with richer information, enhancing their understanding of Braille and its meaning while addressing the challenge of associating text with real-world objects. Additionally, the tool offers intelligent teaching support, streamlining lesson preparation and enabling more effective teacher-student interaction in the classroom.

3.2. Interaction system

The interaction process defines how visually impaired children engage with the assistive tool, forming the core framework of the system (Fig. 3).

First, visually impaired children read aloud from a Braille book they are touching. The system then uses speech recognition technology to analyze the spoken content, and AI-based image generation technology

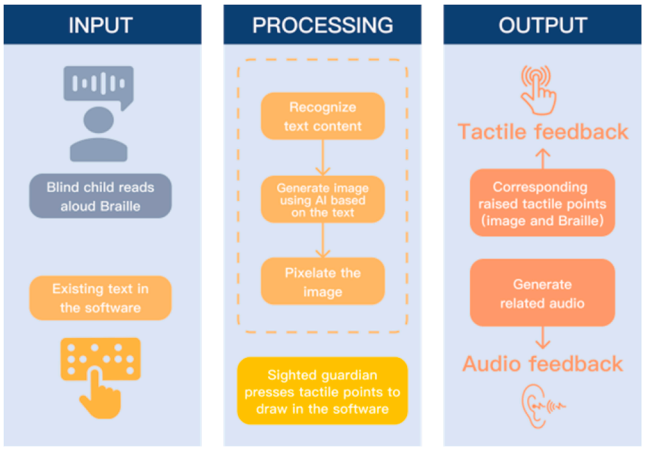


Fig. 3. Product interaction framework.

automatically creates related scene or object images. These images are formatted in black and white, with pixel-style processing to match the hardware's tactile display.

The final output is tailored to the sensory needs of visually impaired children and includes both tactile and auditory outputs. The tactile output consists of raised dots corresponding to the generated image, allowing children to perceive the shapes and contours of the objects they are reading about through touch. The auditory output complements this by providing relevant sound effects and voice narration, enhancing the multidimensional experience of text comprehension.

For example, when a child presses the voice button and reads aloud, "a mountain," and confirms with the enter button, a corresponding scene is generated—a tactile image displaying the contours of a mountain. The child can touch and feel the height and shape of the mountain, while Braille representing "a mountain" is displayed below. Sounds of birds chirping and wind blowing through the mountain trees are played to heighten the sense of realism.

3.3. Hardware design

We divided the touch display into two sections: the upper part displays pixel-based images, while the lower part is reserved for Braille. This design reflects a deep understanding of how visually impaired children learn, allowing them to combine tactile exploration of both images and text for a more comprehensive learning experience. On the right side of the touch screen, control buttons are added, including power, record, stop, page navigation, and volume adjustment buttons. A small speaker is placed in the upper-right corner to support the future implementation of multimodal interaction.

Discussions with teachers from schools for the visually impaired helped finalize the overall size of the product, as well as the dimensions and spacing of each tactile dot. Each dot measures 5 mm x 5 mm, with a 1 mm gap between them. The total size of the product is approximately 360 mm in length, 210 mm in width, and 30 mm in height—comparable to a laptop, making it portable and convenient for class learning scenarios. The tactile image display adopts a raised, positive-relief design, which enhances tactile perception for visually impaired users.

Referring to earlier research on electromagnetic-driven tactile display devices, we employed Arduino, an open-source microcontroller platform widely used for controlling electronic devices, along with a multi-channel relay and micro magnets to create a system capable of controlling the elevation of six magnetic dots. In this system, Arduino is responsible for managing the on-off states of the electromagnetic circuits, enabling the precise raising and lowering of Braille dots and tactile images (Fig. 4).

3.4. Software design

To achieve efficient language-to-image conversion, a software system was developed, which is based on Python and Unity (a cross-platform game engine used for creating both 2D and 3D interactive applications including games and simulations). The system consists of two main components: a Unity-based front-end responsible for user interaction and visual output, and a Python back-end that processes data and executes core functions.

3.4.1. Unity front-end design

The Unity front-end focuses on providing an intuitive and responsive user interface. After reading a Braille text, the user inputs speech via a microphone, engaging interactively in the image generation process. The visual display uses a dynamic grid of blocks, where the height of each block (ranging from 0 to 1) is adjusted based on the generated image, simulating the pixel-like representation of real-world objects. This design mirrors the physical tactile pixel image creation process, enhancing the experience for sighted users and offering them control over visual interaction.

3.4.2. Python back-end architecture

The back-end system, built using Python and the Flask framework, processes the speech data transmitted from the Unity front-end. The system employs Baidu's speech recognition API to convert speech into text, selected for its superior accuracy and performance in processing Chinese. Once converted, the text undergoes custom algorithmic optimization to generate prompts suitable for image creation. These prompts are used to call OpenAI's DALLE-3 model, with specific instructions for "simple, child-friendly, black-and-white pixel-style images." The prompt also generates music that aligns with the theme of the image. The resulting images are then processed using OpenCV (an open-source library used for computer vision tasks, such as image and video processing, enabling machines to understand and analyze visual data) for pixelation, binarization, and size adjustments to ensure seamless integration with the Unity block array.

This system not only demonstrates an efficient speech-to-image conversion process but also integrates auditory feedback through music output. By combining visual, tactile, and auditory elements, it introduces an innovative multimodal interaction model, highlighting the system's broad application potential and promising development prospects (Fig. 5).

4. Methodology

This section outlines the participants, settings, and materials used in the study, followed by an overview of the materials and procedures employed. Lastly, we describe data collection, analysis, and usability

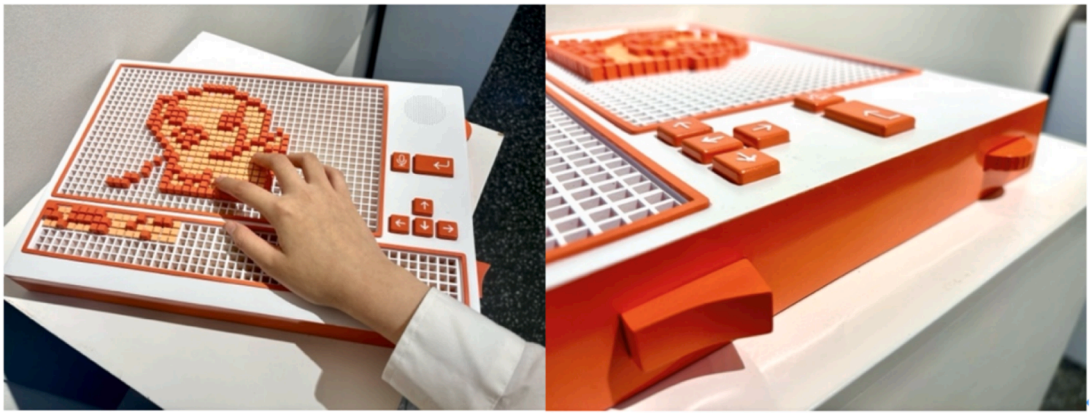


Fig. 4. Product design.

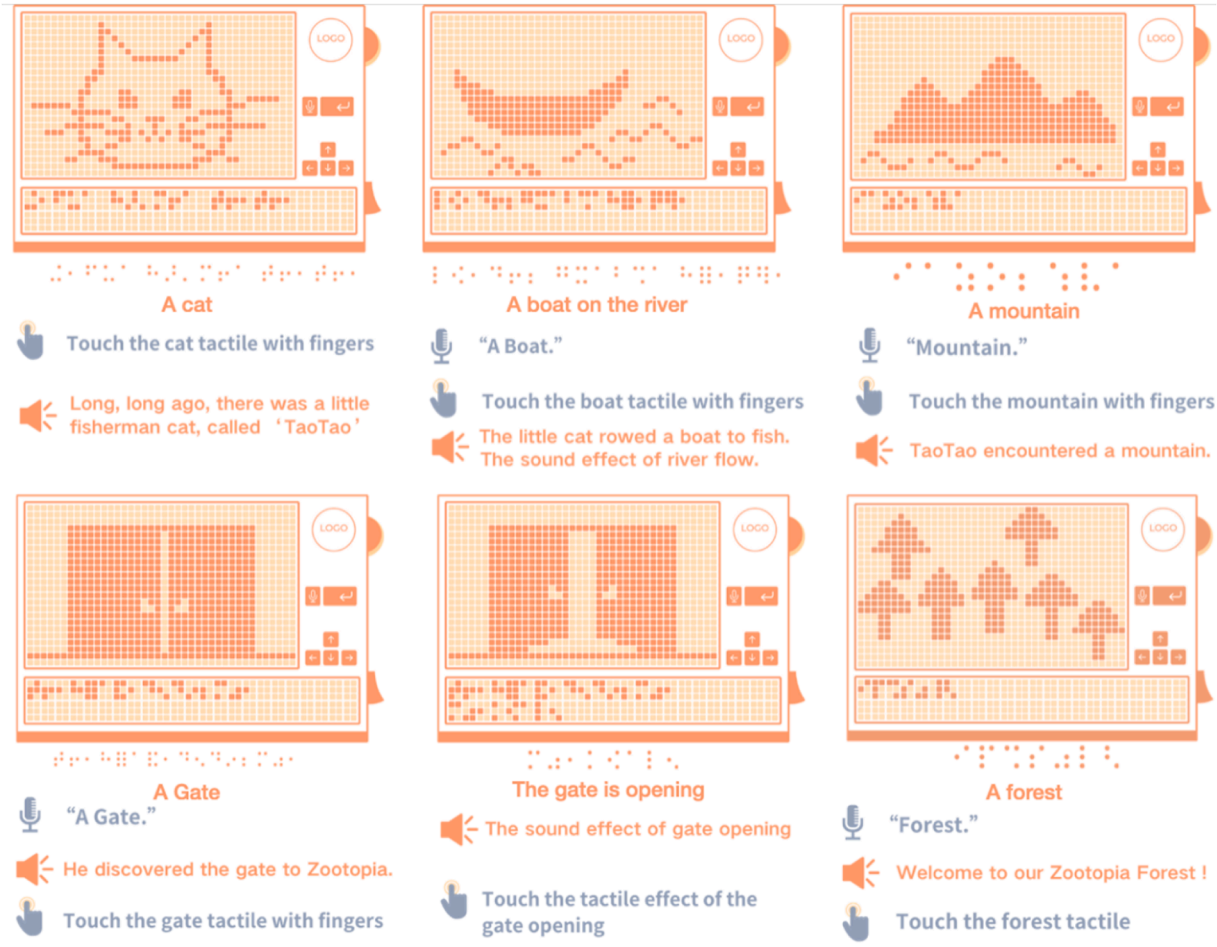


Fig. 5. Example of multimodal interaction of TaleVision.

metrics.

4.1. Participants

The study received approval from the University Ethics Committee (H20230365I). The participants comprised five sighted children (mean age = 9.2, 3 boys and 2 girls) and fourteen visually impaired children (mean age = 9.4, 9 boys and 5 girls), recruited with the assistance of teachers from a school for the blind. All visually impaired participants met the following inclusion criteria: (1) congenital blindness, (2) enrollment in a school for the blind for over one year, (3) blindness not caused by other diseases, and (4) working at age-appropriate cognitive and communication levels. The sighted children were required to demonstrate age-appropriate cognitive and communication abilities. Informed consent was reviewed by the participants' guardians, who were assured that participants could rest or withdraw at any time. After addressing all questions from the guardians or participants, the guardians signed the consent forms. Meanwhile, another researcher set up the equipment.

The purpose of testing with sighted children was to determine whether the generated images aligned with their cognitive understanding of objects. Following their positive feedback, the usability testing with visually impaired children commenced. Participants were observed in two locations: fourteen visually impaired children at a school for the blind, and five sighted children at a local elementary school. Each participant received a $20 gift card as compensation. Data collection took place between April and May 2024.

4.2. Materials

The primary tool used in this study was the "TaleVision" product, designed to generate tactile images and corresponding sound effects by large language models (LLMs). The system was programmed to produce touchable representations of various objects, animals, and simple scenes, accompanied by audio descriptions and sound effects to provide a multimodal sensory experience.

4.3. Experimental procedure

The experiment comprised five stages (Fig. 6). Following the principle that visually impaired children are capable participants, we communicated directly with them rather than their guardians, using age-appropriate language. Various options were provided to familiarize the children with the equipment before obtaining their consent to participate.

In the pre-experiment phase (Part A), sighted participants received a brief introduction to the product and its operation. First, a two-minute subtitled video explaining the product's usage and the tasks involved was shown. Participants then engaged in a blindfolded simulation of the learning process of visually impaired children, interacting with the hardware to experience the product's overall size, material, and functionality. They touched the display image of a "puppy," speculating about the object on the screen, aided by audio cues. Afterwards, they removed their blindfolds and used the system's voice input and image generation functions.

In Part B, visually impaired participants received guidance from teachers, along with a brief introduction and operational instructions.

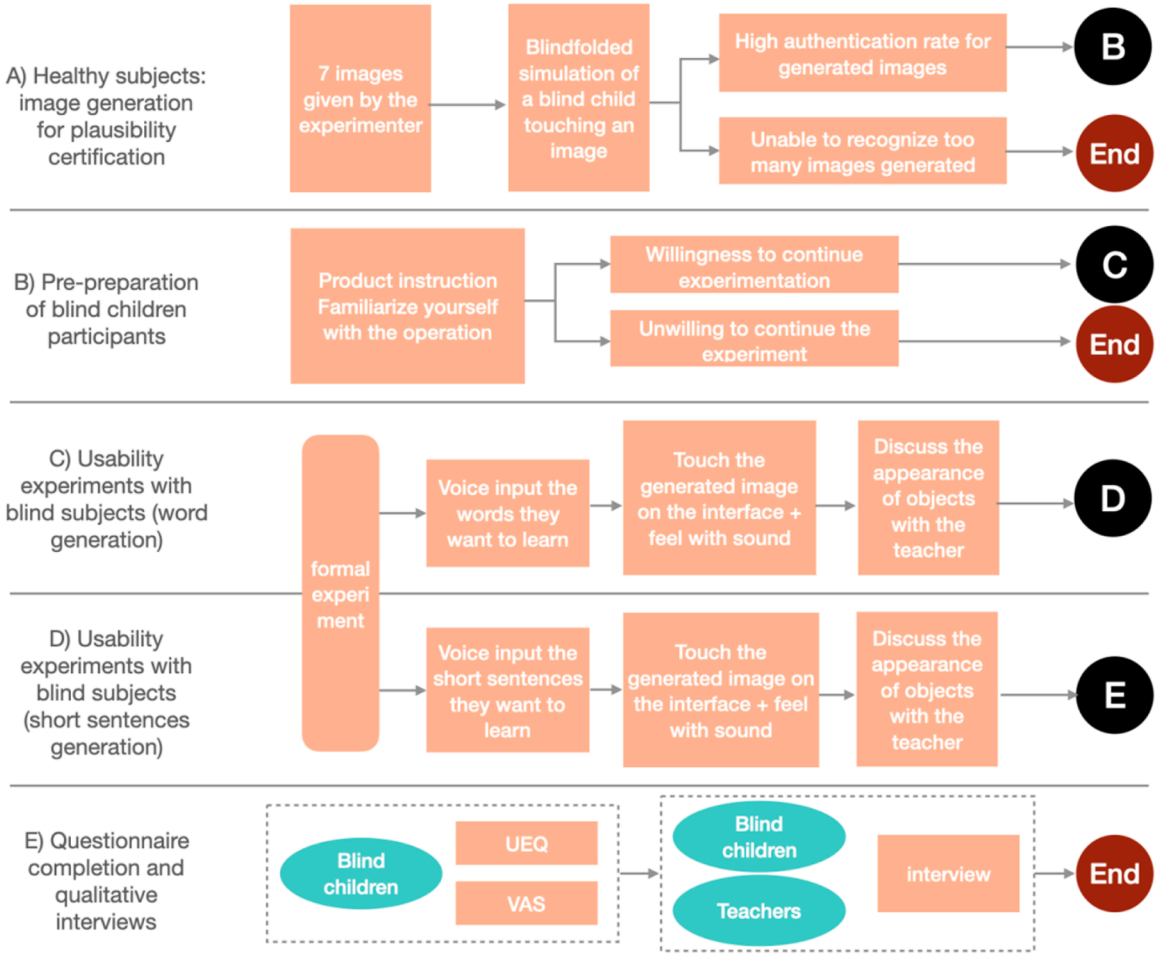


Fig. 6. Experimental flow.

Each visually impaired participant had five minutes to familiarize themselves with the product's buttons and interaction methods. Researchers explained what participants would experience and perform during the study—such as voice input of words or sentences, pressing the image generation button, and interacting with the tactile display. During the pre-test, participants could ask questions, and they were free to withdraw from the experiment for any reason.

First, voice the words from your textbook into TaleVision. Then, explore the touchable images generated and listen to the corresponding sound effects. Participants were asked to describe the objects they perceived and provide feedback on their understanding of the images. Teachers guided the participants through specific tasks and were asked to comment on how the tool supported or challenged their instructional methods. Researchers recorded the participants' behaviors, operational steps, and reasons for task failures. Participants were allowed to take breaks if needed, and researchers invited them to continue when ready.

In Part D, visually impaired participants were instructed to input complex descriptive phrases via voice commands, such as "a pavilion on a mountain" or "a boat passing by several mountains on the water." They then followed the same procedure as in Part C. Researchers documented their performance, behaviors, and any operational challenges.

After completing the experiment (Part E), participants filled out a user experience rating scale and satisfaction survey, specifically designed to assess engagement, satisfaction, and ease of use. Visually impaired students were assisted in orally completing the forms. Finally, structured interviews were conducted, asking participants questions such as: "Did you enjoy using this product?" "What improvements would you suggest?" and "At what point did you experience difficulties?"

4.4. Data and metrics

The study gathered data from the pre-test conducted with sighted children. Specifically, we presented seven commonly taught words in class: "dog," "cat," "flower," "mountain," "child," "tree," and "boat." The sighted children, blindfolded, were asked to touch the images displayed by the product and guess the objects represented. Their accuracy rates were recorded.

Data collected from the visually impaired children included: (a) quantitative data, such as participant satisfaction and user experience, and (b) qualitative insights derived from researcher observations. Due to the sensitivity of participant identities and school policies, researchers manually recorded events in digital form, as video or photographic documentation was not permitted.

Our analysis focused on four key metrics: (a) participant satisfaction with the product, (b) their user experience, (c) overall performance during product interaction, and (d) reasons for interruptions or task failures. These metrics are described in detail below:

4.4.1. Participant satisfaction

At the end of the experiment, subjective satisfaction was assessed using a modified Visual Analogue Scale (VAS) tailored for visually impaired children. The VAS is a simplified psychometric system designed to collect individual perceptions of specific experiences. This adjusted scale included three smiley faces, each representing different satisfaction levels: a sad red face for "not at all," a neutral yellow face for "somewhat," and a happy green face for "very much liked" (Wong and Baker, 2001) (Fig. 7). The faces, with high-contrast colors, were easily



Fig. 7. VAS scale (Visually Impaired Child Friendly Version).

perceptible for visually impaired children, and each had a tactile mouth to aid comprehension. Touching the mouth and its orientation (downturned, straight, or upturned) helped children understand the corresponding expression. After being shown the smiley faces one by one and allowed time to explore them, participants were asked, "Did you enjoy interacting with this product?" Children provided responses by either touching or pointing at a face, or by answering verbally. Researchers recorded their responses at the conclusion of each session.

4.4.2. User experience

The user experience of visually impaired participants was assessed using the User Experience Questionnaire (UEQ) (Schrepp, 2015). Researchers verbally conveyed the questions and options to ensure clarity and accuracy in expressing participants' feelings about the product. The six dimensions of the UEQ (Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty) were retained, but the presentation was adjusted to align more closely with the participants' actual interactions. Researchers paid particular attention to the participants' reactions to tactile feedback, audio cues, and the overall interaction process, ensuring that these aspects were reflected in the UEQ scores. A 7-point Likert scale was used for each dimension, defined as follows:

- Attractiveness: Measures participants' overall satisfaction and emotional response to the product.
- Perspicuity: Evaluates the product's ease of use and clarity in its functionality.
- Efficiency: Assesses how effectively participants could complete tasks using the product.
- Dependability: Measures the product's predictability and the level of control participants felt during use.
- Stimulation: Reflects the enjoyment and positive emotions elicited by the product.
- Novelty: Assesses whether the product provided a unique and innovative experience.

4.4.3. Task performance

This metric assessed participants' operational skills and task completion rate during product use. The research team observed and recorded the number of tasks completed within 30 min, interruptions during tasks, and instances where participants disagreed with the generated images. The task performance of visually impaired participants was indicative of their ability to identify objects using the product and provided insight into its applicability in educational settings.

4.4.4. Interaction performance

We observed and recorded conversations between visually impaired children and their teachers during interactions. These interactions were categorized into 1) Interactive operations; 2) Judgments about images; 3) Characterizing objects and expressing ideas. Given the potential technical limitations and the specific needs of the target population, it was anticipated that participants might face challenges during product use. Therefore, the study also documented instances where learning was interrupted or tasks were unsuccessful. Reasons for these interruptions were classified into (1) difficulties in tactile recognition, (2) cognitive overload from multimodal inputs, and (3) excessive complexity of the product's feedback. This metric aimed to identify barriers to the learning process and provide insights for product optimization, such as simplifying image complexity or adjusting the frequency of audio prompts.

We employed an abductive content analysis method to analyze the recorded data in NVivo (Version 12; QSR International) (Graneheim et al., 2017; Timmermans and Tavory, 2012). In each group, the first author coded the records, aiming to understand how visually impaired children, with teacher assistance, engaged in learning and behavior. This process involved reviewing the records line by line, extracting relevant excerpts, and identifying codes (children=14, teacher=2), with a focus on what visually impaired children attempted to accomplish and how they did so. Subsequently, sub-themes related to the methodology were identified ($n = 6$). The authors then collaboratively refined these codes and sub-themes, which were organized into broader themes ($n = 3$) concerning the interaction between visually impaired children and teachers within the three main components of the product. Examples of data extraction are provided in Appendix File 1.

5. Results

5.1. Product testing with sighted children

Preliminary data from five sighted children participating in the product test revealed an overall accuracy rate of 65.71 % in identifying tactile images. The highest recognition rates were for "flower" and "boat," both at 100 %, while "tree " and "mountain" were recognized 60 % of the time, and "dog " only 40 %. More complex figures, such as "child," had a significantly lower recognition rate of 20 %, whereas simpler natural objects like "flower" and "boat" had higher accuracy. This suggests that simple line-drawn images are more effective for tactile recognition, whereas more intricate images may require further optimization to improve identification accuracy among visually impaired children. Given that prior literature indicates sighted children are generally less proficient in tactile perception compared to their visually impaired counterparts, we hypothesize that visually impaired children may possess greater tactile recognition abilities for simpler images. Based on these insights, we proceeded with a formal usability experiment involving visually impaired children.

5.2. Usability testing with visually impaired children

5.2.1. Quantitative analysis of usability testing

Fourteen children participated in a 30 min interaction with the product, and the results indicate a high level of overall satisfaction. The average satisfaction score among the participants was 2.6 (SD = 0.50), with 60 % expressing very high satisfaction. In terms of task performance, children successfully completed an average of 2.9 (SD = 0.77) tasks involving image generation from individual words. However, their performance in sentence-based tasks was limited, with most participants completing only one sentence task, suggesting that large language models may not yet be suitable for generating complex images for visually impaired children. Regarding errors, most children experienced 2–4 task interruptions, reflecting the adequacy of the product's button configuration in maintaining their focus. A few children experienced higher interruption rates, primarily due to unfamiliarity with the directional buttons. Additionally, several children exhibited instances of image misidentification, with the highest misidentification rate being two instances per child. For example, some children mistook "mountain" for "cloud".

An analysis of the relationship between satisfaction and interruptions reveals an inverse correlation between these two factors. Children who reported higher satisfaction (e.g., Children 5, 6, 10, and 13) generally experienced fewer interruptions, suggesting that smoother task flow with minimal disruptions positively influenced their overall experience. For instance, Child 6, who encountered no interruptions, exhibited a high satisfaction score and successfully completed multiple

word tasks, reinforcing that reducing interface-related disruptions enhances user satisfaction. Conversely, participants with higher interruption rates (e.g., Children 4, 9, and 12) reported lower satisfaction, implying that frequent task disruptions due to unfamiliarity with controls diminished their engagement and positive perception. Notably, Child 4 experienced the highest number of interruptions (6) and reported lower satisfaction, emphasizing the need for more intuitive tactile navigation. A similar pattern emerges when examining task completion. Children who completed more word tasks (e.g., Children 1, 5, 6, and 13) tended to report higher satisfaction, while those with fewer completed tasks (e.g., Children 4, 9, and 12) demonstrated lower satisfaction, highlighting the connection between successful task execution and positive user experience. Interestingly, sentence completion rates remained consistently low across participants, regardless of satisfaction levels, suggesting that the complexity of sentence-based tasks remains a significant challenge (Fig. 8).

These findings suggest that improving interface familiarity through training, introducing adaptive tactile prompts, and providing personalized interaction pacing could reduce interruptions and further enhance satisfaction in future iterations of the product.

5.2.2. Experience feedback

The fourteen participants provided comprehensive feedback based on six dimensions of the User Experience Questionnaire (UEQ) (Fig. 9). The product received particularly high scores for attractiveness (2.0, SD = 0.96), clarity (1.8, SD = 0.70), and novelty (2.4, SD = 0.50), reflecting strong user satisfaction and an appreciation for the product's innovative features. Participants highlighted that the interactive combination of touchable images and auditory feedback made the learning process engaging and provided a novel experience compared to traditional Braille learning tools.

However, the efficiency score was relatively low (1.1, SD = 0.77), indicating that participants experienced delays and operational complexity when interacting with the system. Some participants reported that the response time for generating tactile images was longer than expected, which disrupted the learning flow. Suggestions for improvement included optimizing the system's processing speed and simplifying the interface to facilitate faster, more intuitive interactions.

The dependability score (1.8, SD = 0.80) suggests moderate user confidence in the system's performance consistency. Participants noted occasional errors in tactile image generation and the need for more precise haptic representations. To address these concerns, future enhancements should focus on improving system reliability through more refined 3D modeling and robust error-checking mechanisms.

While the product's novelty received the highest score (2.4, SD = 0.50), a few participants suggested that the range of touchable images could be expanded to include more complex shapes and real-world objects, further enriching the learning experience. The stimulation score (1.8, SD = 0.63) indicates that users found the product engaging, although some suggested adding more interactive features, such as multi-sensory exercises, to sustain long-term engagement.

In comparison to the benchmark derived from the UEQ database—which includes 163 applications across various categories, such as business software, online services, and mobile applications—the product performed above average in attractiveness and novelty. However, it fell short in efficiency and dependability, highlighting areas where technical optimization and user interface enhancements are needed. These findings underscore the product's potential while identifying specific aspects requiring improvement to better serve the learning needs of visually impaired children.

5.2.3. Usability experiment: qualitative analysis

1. Language Visualization

   Overview: Language visualization is an innovative technique that uses large language models and generative AI technologies to transform textual descriptions into tactile images for visually impaired children. In this process, the system generates corresponding images based on the text, with the complexity and level of detail directly influencing the children's comprehension and learning outcomes. The children primarily expressed two points: "What benefits do I find in the generated image?" and "Why do I like or dislike this image?"

   Uncertainty in Generation

   "I tried inputting the word 'puppy' twice... but the resulting images were different each time—one simple and the other complex—which caused me some confusion. When generating images from sentences, the results are often highly complex, making them difficult for me to understand."

   This statement highlights a key challenge in applying text-to-image models for visually impaired children—the need for fine-tuning to reduce variability in generated images. Simple outlines and structures are easier for the children to recognize, whereas complex scenes or excessive details may cause confusion. Additionally, when voice inputs include multiple nouns or when more than two objects need to be generated, the resulting images often become too pixelated or abstract for clear interpretation.

   Diversification of objects

   "It's fascinating... no matter what word I say, it responds with an image. For instance, when it generated a monster, I got my first tangible concept of what it looked like... I can share these moments with my family."

   The richness of generated images is also central to enhancing the learning experience. By allowing visually impaired children to explore and interpret these images freely, the system encourages creativity and fosters a more active role in understanding content. Offering varied inputs can help generate a broader range of image scenarios, catering to diverse learning needs.

2. Human- computer Interaction Process

   Overview: During interaction, visually impaired children focus on

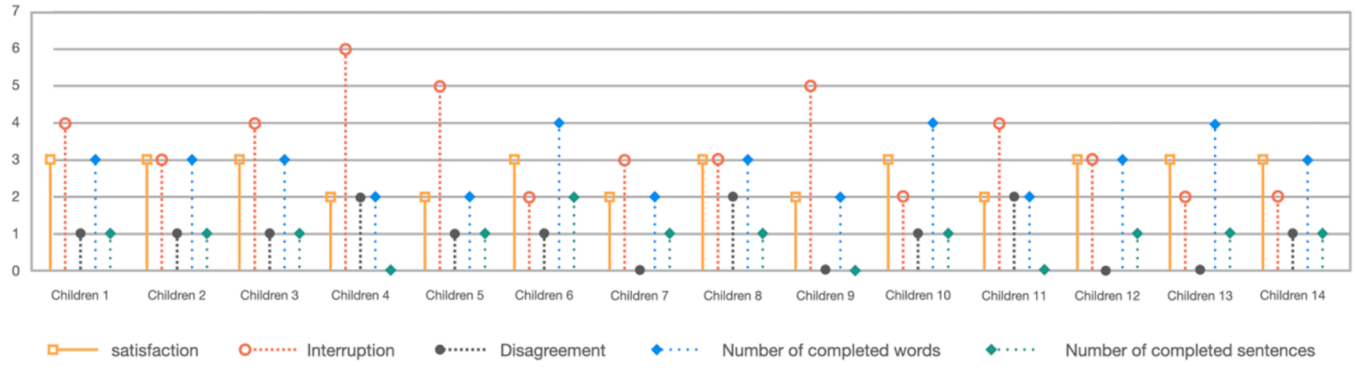


Fig. 8. Quantitative analysis of data from usability experiments.

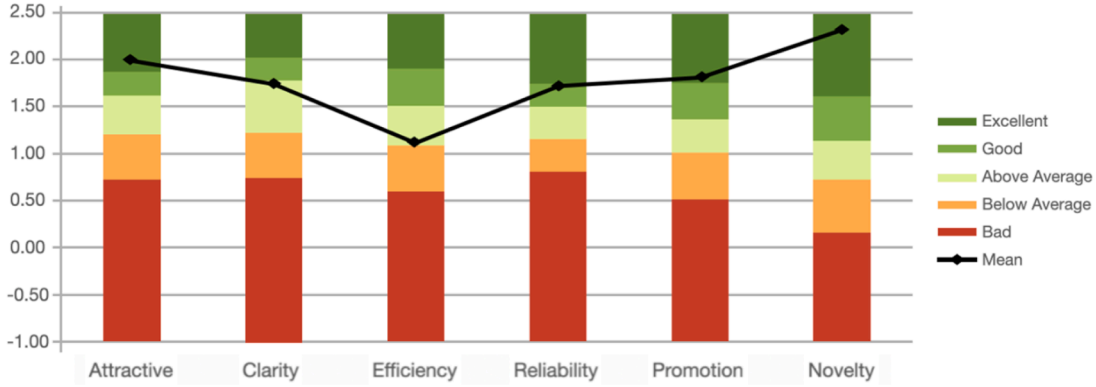


Fig. 9. Data from the subject user experience scale.

constructing their own technological practices. However, due to varying levels of technical literacy, we found that their product use practices were inconsistent. Many teachers attributed this disparity to the children's lack of exposure to electronic devices in their daily lives. We observed that the children developed different methods of interacting with the product through tactile and verbal feedback. The following sections explore their varying attitudes toward product design.

Simplification

"The five buttons are very helpful... it keeps everything organized because I can't quickly understand the function of each one... AI is unfamiliar to me. We need time to comprehend it."

By optimizing system responsiveness and feedback mechanisms, the product can more quickly perceive and respond to the children's actions, ensuring they receive timely feedback and reducing frustration. Standardizing operational procedures is a key aspect of simplification, streamlining complex steps so the children can quickly grasp and remember the process, ultimately easing the cognitive load required to learn new devices.

Standardization

"It's important that this system benefits our care services... but if I want to revisit an image... once it's gone, it's gone."

This statement underscores the need to establish a unified design framework that ensures new products are compatible with the children's familiar usage patterns. Such standardization not only facilitates seamless transitions between different devices but also improves their overall user experience with digital products.

3. Teacher-Student Interaction in Learning

Overview: During the teacher-student interaction phase, visually impaired children use the product with the guidance of their teachers, resembling a traditional classroom setting. Their tasks include touching images, discussing their appearance with the teacher, and articulating their understanding of the images. Teachers guide the students in recognizing object shapes and simultaneously nurture their independent learning and expression abilities. We observed that through teacher-guided interaction, visually impaired children deepen their understanding of the learning content by integrating external information and validating internal knowledge.

External Information Supplementation

"I find that exploring object features with students helps them better understand... I guide them to think about what certain parts are and what they're used for, then offer explanations afterward... these interactions prompt the children to ask questions, aiding their memory retention... the sound effects are engaging too, keeping them focused on the task."

During teacher-student interactions, visually impaired children receive additional details about the objects depicted in the images. For instance, when touching an image of a tree, the teacher might explain its species or the shape of its leaves. This additional information helps the children build a more complete cognitive picture, compensating for the limitations of tactile feedback alone. Furthermore, combining auditory descriptions or relevant sound effects enhances the clarity of the information. For example, when touching an image of a cat, the teacher might play a recording of a cat's meow. This multi-sensory approach reinforces recognition and memory retention, significantly improving learning efficiency.

Internal Knowledge Validation

"When I touched the umbrella image, I never realized it was shaped like that... I usually only touch folded umbrellas or stand beneath open ones... the teacher said this image shows the umbrella from the side... it gave me a new perspective I hadn't considered before."

This statement illustrates how visually impaired children deepen their understanding by connecting the learning content to real-life experiences. Teachers guide them to recall past encounters with the object, such as touching real items or participating in related activities. This method enables the children to link abstract tactile images with actual objects and experiences, fostering knowledge internalization. Additionally, the children are encouraged to translate what they have learned into their own language and understanding, thereby expressing their cognitive grasp of the subject matter.

6. Discussion and recommendations

6.1. Discussion of results

The objective of this study was to evaluate the usability of a multimodal perception integration product, based on large-scale models, for enhancing image cognition among visually impaired children. Through both qualitative and quantitative analyses, we discovered that the majority of participants effectively utilized this innovative interactive product to recognize objects in their environment. Specifically, 60 % of the visually impaired children expressed satisfaction with the product, which demonstrated high levels of novelty and attractiveness within the user experience framework. Consequently, we addressed the following research questions:

RQ1: Can large-scale models serve as a technical foundation for graphic cognition educational products tailored to visually impaired children?

Large-scale models exhibit substantial potential in empowering graphic cognition educational products. The images generated by these models facilitate the understanding of abstract concepts among visually impaired children, particularly through a multimodal perception system that integrates tactile and auditory stimuli. This integration enables children to better perceive and comprehend image content that is otherwise inaccessible (Zhang et al., 2017). However, the complexity of

the generated images poses a significant challenge. Our observations indicated that large-scale models struggle with visualizing complex short phrases, resulting in cognitive confusion due to the excessive aggregation of elements within the images (Tekli et al., 2018). Controlling image complexity appropriately can mitigate confusion during the cognitive process. Additionally, fine-tuning these models is essential to ensure that the generated images are better aligned with the practical needs of visually impaired children. In summary, the inherent flexibility of large-scale model technology also allows for diverse input types, fostering children's imagination and facilitating a richer image generation experience.

Compared to existing tactile display technologies, such as the APH Monarch and Dot Pad, our approach using large-scale models introduces greater flexibility and adaptability. While the Monarch relies on pre-existing Braille books and the Dot Pad requires pre-defined visual input (Hu et al., 2022; Rao et al., 2021), our product can dynamically generate tactile representations from diverse input sources, reducing the reliance on pre-programmed content. This dynamic capability better accommodates the spontaneous learning needs of visually impaired children. However, it is also essential to address the challenge of image complexity and improve model fine-tuning for more accurate and comprehensible outputs.

RQ2: How do visually impaired children collaborate with teachers while using the product, and what is the manifestations of their interaction?

The collaboration between visually impaired children and teachers in utilizing the product constitutes a process of information complementarity and knowledge co-creation (Gadiraju, 2023). Teachers provide supplementary information, such as detailed explanations and contextual backgrounds of the images, while leveraging the multimodal perception system (tactile and auditory) to enhance the learning experience. This interaction transcends mere one-way information transmission; it involves teachers guiding the children's perceptions and facilitating the transformation of real-world objects into cognitive understanding. Teachers help overcome the limitations of tactile cognition by offering external information, whereas children deepen their comprehension through continuous interaction and feedback, ultimately constructing a unique knowledge system during the product usage (Cuturi et al., 2016). The essence of this interaction lies in the integration of external information with internal knowledge validation, thereby progressively unifying sensory experiences with cognitive processes.

Unlike static materials that rely on fixed content delivery, our product allows for real-time adaptation and teacher-guided exploration. This dynamic process fosters a more personalized learning environment where both tactile and auditory channels are integrated. Nevertheless, further studies are required to examine how these collaborative interactions evolve over longer periods and to identify best practices for optimizing teacher-child cooperation.

RQ3: What are the experiences, views, and preferences of visually impaired children and teachers regarding the touch interface input method?

Visually impaired children and teachers exhibit distinct experiences and preferences concerning the touch interface input method. Simplification emerges as a core element in enhancing the product experience, particularly for visually impaired children. Enhancing the responsiveness of the touch interface and standardizing operational procedures are crucial for improving usability (Kieras, 1988; Xu et al., 2018). Concurrently, standardized design ensures that the product aligns with the interaction methods of digital devices commonly used by visually impaired children, thereby making operations more intuitive and consistent (Sánchez, 2008). Visually impaired children prefer touch screen inputs that are easy to operate and highly responsive, whereas teachers prioritize design strategies that adapt to the children's usage habits and provide adequate informational feedback. This dual perspective on touch screen input reflects the children's need for convenient operation and the teachers' emphasis on product operability and effective interaction methods.

Compared to other smart educational tools, such as speech synthesis devices and screen readers (Radecki et al., 2020; Maćkowski et al., 2023), our touch-based interface offers a multisensory experience that combines tactile exploration with auditory feedback. While screen readers primarily provide auditory access and lack tangible engagement, our product enhances the sensory interaction by allowing children to actively manipulate and interpret image content. This approach aligns with prior research emphasizing the importance of sensory integration in fostering cognitive development (Mascle et al., 2022; Slade et al., 2021). However, future iterations should continue refining the touch interface to ensure consistency and accessibility across diverse educational contexts, addressing both the children's need for simplicity and the teachers' need for efficient instructional tools.

6.2. Design recommendations

Based on the findings of this study, we propose several design recommendations:

"Systematic Design for Learning, Not Just One-time Interactions"

When designing products for visually impaired children, it is imperative to move beyond one-time interactive experiences and develop a comprehensive learning system. During the experimental phase, visually impaired children indicated an inability to revisit previously completed images, suggesting that simplicity should not be achieved by reducing functionality (Markopoulos et al., 2008). A complete system should encompass features relevant to the learning process, such as saving and recording previously learned images and content for future review and deeper understanding. This approach enables visually impaired children to build long-term memory through continuous learning rather than relying solely on isolated experiences.

"Necessity of Fine-tuning Large-scale Models"

While large language models demonstrate significant potential in image generation, their default outputs may not always align with the specific learning needs of visually impaired children. Therefore, fine-tuning these models to address the unique requirements of this demographic is essential (Ruiz et al., 2023). Training the models with extensive annotated image data tailored for visually impaired children can enhance the models' understanding of the children's needs, particularly concerning contour clarity and tactile feedback recognizability. Additionally, implementing adaptive algorithms that continuously adjust image details based on feedback from visually impaired children ensures the generation of images that are more educationally meaningful and appropriate for their cognitive development stages (Zeng et al., 2024).

In our study, we identified several key failure points, including high interruption rates due to unfamiliarity with the task interface, image misidentifications such as mistaking a "mountain" for a "cloud," and limitations in tasks that require more complex contextual understanding. These challenges underscore areas where the model's performance could be improved.

To address these issues, we propose a focused fine-tuning approach that specifically targets these failure cases. First, we recommend using a dataset tailored to these problems, incorporating multimodal data with child-friendly descriptions and clear object visualizations. Relevant datasets include *COCO-Captions*, which provides diverse object-text pairs, and *Conceptual Captions*, which includes broader object and

scene descriptions. Additionally, for improving accessibility, we suggest leveraging datasets like *VizWiz*, which is designed for visually impaired users and emphasizes clear, easily recognizable object shapes. By manually marking cases of misidentification and expanding the dataset with ambiguous or unclear examples, we can improve the model's ability to disambiguate between similar concepts. Furthermore, capturing user interaction logs will help us understand where children encounter difficulties, allowing the model to be adjusted for more accurate guidance and error correction. Additionally, we plan to explore tactile data resources such as the *TGIL*repository, which offers a collection of 2385 tactile graphics templates that can be used to create graphics or be embossed for classroom use. This resource could provide a foundation for further enhancing accessibility through touch-based learning.

In terms of fine-tuning methodology, we aim to use data augmentation techniques to enhance the model's ability to handle ambiguous prompts, as well as contrastive fine-tuning to differentiate between visually similar objects. We also intend to implement instructional fine-tuning, where structured prompts and responses help guide children step by step through tasks, addressing difficulties as they arise. Lastly, by using interaction logs, we can reinforce the model's ability to detect when additional guidance is needed, reducing instances of navigation errors. While a fully developed cross-modal tactile-visual dataset specifically designed for visually impaired children is not yet widely available, we plan to explore datasets like the *Natural Scene Braille Image Dataset* (which contains images with Braille labels) and *GelSight*, a dataset developed by MIT researchers using tactile sensors to categorize textures like fabric, wood, and sandpaper. Additionally, the *Touch and Go* dataset, which supports multimodal tactile-visual learning, could help in improving touch-based interactions within the system. Through this targeted approach and by leveraging these resources, we are confident that we can improve the model's performance and its ability to function effectively in educational settings.

"Transition from 2D to 3D for Enhanced Object Cognition"

Interactions with visually impaired children and visually impaired children revealed that while some can recognize flat, cartoon-like shapes, they often struggle to associate these shapes with real-life objects, such as dogs or cats. This indicates the limitations of 2D images in conveying complex and realistic objects. A key design recommendation is to develop images with 3D effects, which can facilitate easier and more direct associations between tactile shapes and familiar real-life objects for visually impaired children (Sánchez and Sáenz, 2006). Furthermore, visually impaired users expressed a preference for finer and smaller surface textures on the interactive interface, as heightened tactile sensitivity makes overly large grains detrimental to their perception of details. Adopting finer grain designs can provide richer tactile information, enabling users to discern subtler differences and accurately identify images.

Future iterations of *TaleVision* will employ advanced AI-driven techniques to generate more detailed and accessible tactile outputs. We plan to utilize AI-based 3D model generation tools (e.g., Hunyuan3D, Nerf, or Depth Anything,) to create corresponding 3D models from 2D images (Fig. 10). By sampling and stratifying the height values across the 3D model surface, we will reconstruct the surface morphology through variations in block heights, allowing for a more accurate tactile representation. Additionally, we will explore monocular depth estimation models (e.g., DPT and MiDaS) to extract depth maps directly from 2D images, enabling the tactile display to represent multi-layered object contours. This dual approach, combining 3D model generation and depth estimation, will allow for adaptive scaling and enhanced tactile feedback. To optimize user experience, we will also implement a hybrid 2.5D model that emphasizes critical object features while simplifying non-essential areas, ensuring a balance between tactile clarity and structural complexity. Furthermore, we plan to introduce dynamic height control, allowing users or instructors to customize the tactile resolution based on individual sensory needs. These enhancements will provide a more immersive and precise learning experience, facilitating a clearer understanding of complex visual concepts for visually impaired children while maintaining adaptability to diverse educational contexts.

"Essential Role of Teacher Involvement in Learning"

Another critical insight is the indispensable role of teacher involvement in the learning process of visually impaired children. Teachers are not merely auxiliary support but act as central guides, providing personalized instruction and immediate feedback to facilitate deeper understanding of learning content (Mountapmbeme and Ludi, 2021). This guidance is particularly vital when using multimodal sensory interaction products, where visually impaired children rely on tactile and auditory senses in lieu of vision. Timely intervention by teachers can significantly enhance learning outcomes (Rosas et al., 1997). However, without teacher guidance, the effectiveness of the product in promoting autonomous learning diminishes considerably. Observations from

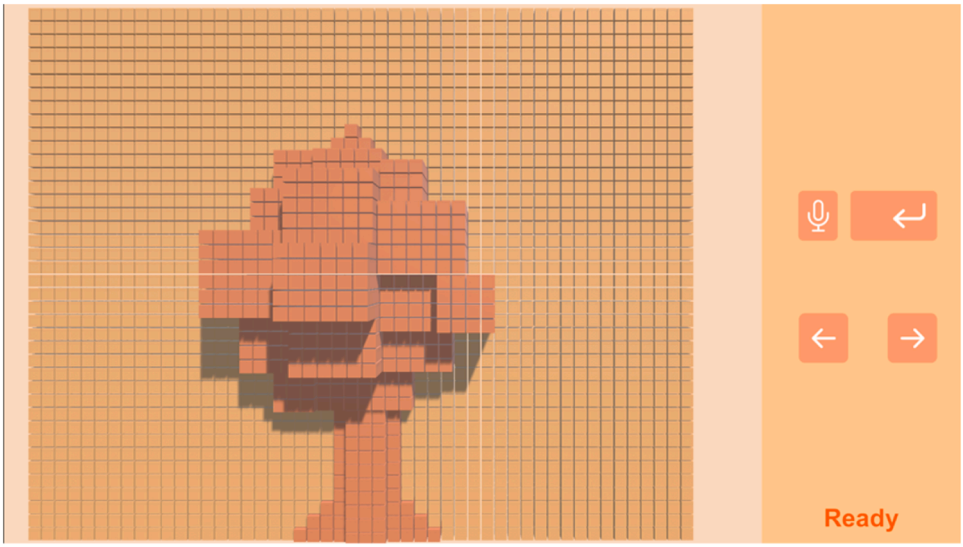


Fig. 10. 3D "tree" display in the unity interface.

testing and usage scenarios indicated that visually impaired children often experienced confusion and a slower learning pace in the absence of teacher guidance. This underscores the necessity of designing additional features and tools that support teacher involvement, such as comprehensive training materials and detailed product usage guides, ensuring that teachers can proficiently utilize the product to assist visually impaired children in their learning journey (Niwagaba, 2014).

7. Limitations

Despite its contributions, this study has several limitations. First, due to the specialized nature of the participants, our sample size was limited to 14 visually impaired children. However, from a usability perspective, a sample of this size is sufficient to identify the majority of usability issues (Nielsen and Landauer, 1993). Second, institutional policies at the school for the visually impaired restricted the use of video or photographic recordings during the experiments. This limitation hindered the capture and analysis of detailed participant behavior and reactions, particularly non-verbal communication cues. Digital records alone may have omitted crucial contextual information. Third, while the images generated by the large model were fine-tuned for the study, there were still variations in image quality that could not be fully mitigated. Future research will focus on more specialized model training for generating images tailored to the needs of visually impaired children. Additionally, the backend system experienced delays, with image generation times for a single word ranging from 10 to 15 s, which may have contributed to user disengagement.

Finally, this study focused exclusively on short-term user experiences and learning outcomes. We did not assess the long-term cognitive development or Braille literacy improvements that could result from extended product use. Future research should employ randomized controlled trials to track the progression of visually impaired children's image recognition abilities, supported by quantitative data to verify the product's efficacy.

8. Conclusions and future work

This study demonstrates that integrating multimodal perception into assistive learning tools can enhance visually impaired children's ability to associate tactile and auditory inputs with Braille. The findings indicate that participants were generally able to grasp the connection between generated tactile graphics and verbal descriptions, yet several key challenges emerged. Issues such as the complexity of AI-generated images, the precision of tactile representation, and the role of teacher guidance significantly influenced usability and learning outcomes. Although the children exhibited strong engagement with the system, frequent interruptions and operational difficulties highlighted areas requiring further refinement.

A critical insight from this study is that while multimodal integration enhances learning, the effectiveness of tactile representations remains contingent on the clarity of image generation and the adaptability of the system to different user needs. The reliance on teacher facilitation also underscores the necessity of structured instructional support. In this regard, developing a more adaptive learning platform—one that allows children to review previously learned content and interact with a personalized curriculum—could substantially improve learning continuity and retention.

Future work should prioritize several key areas. First, a more systematic learning platform is needed—one that allows visually impaired children to save and review previously learned content rather than being limited to one-time usage. Second, to address the precision of image generation, further fine-tuning of the model and more responsive feedback mechanisms should be explored to improve recognition accuracy. Furthermore, the involvement of teachers remains indispensable in the learning process for visually impaired children. Developing a standardized teacher support system is essential to facilitate their guidance in product usage. Lastly, large-scale, long-term studies should be conducted to evaluate the impact of this product on the cognitive development and learning outcomes of visually impaired children.

CRediT authorship contribution statement

Hantian Wu: Writing – original draft, Visualization, Software, Investigation, Conceptualization. **Hongyi Yang:** Writing – review & editing, Project administration. **Fangyuan Chang:** Writing – review & editing, Supervision, Methodology. **Dian Zhu:** Writing – review & editing, Visualization, Methodology. **Zhao Liu:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper. They have no financial, personal, or professional relationships that could be construed as influencing the research presented in this work.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijhcs.2025.103525.

Data availability

Data will be made available on request.

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